

AVIATION FATALITY

DATA INSIGHTS USING PYTHON AND TABLEAU

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09/09/24

INTRODUCTION

- I am here to present key insights from aviation fatality data.
- This will help us understand accident patterns and make informed safety decisions.

PURPOSE OF THE DASHBOARD:

- The dashboard analyzes aviation fatalities to identify trends and areas for improvement.
- This analysis will shed light on the areas requiring attention for better aviation safety.





Data overview

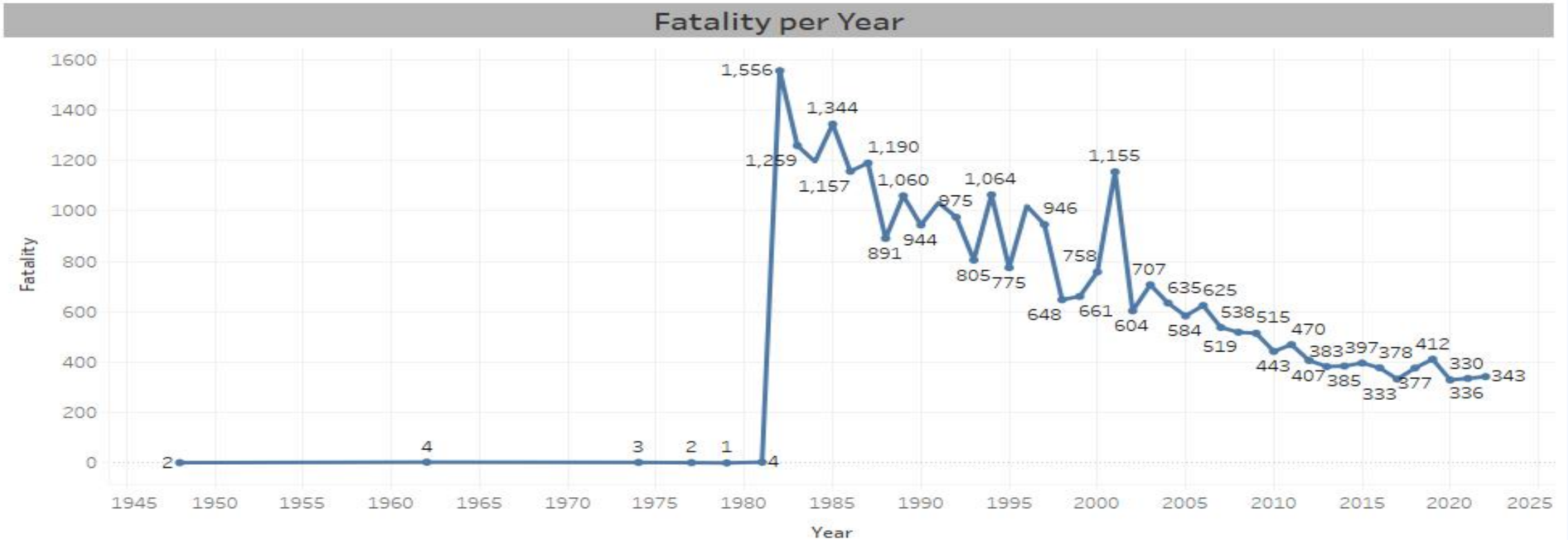
Source of Data:

- The dashboard covers global aviation fatalities, focusing on key factors like time, location, and causes.

Key Metrics:

- The number of fatalities in different time periods.
- Geographic distribution of accidents.
- Patterns in fatalities across regions and airlines.

Fatalities by year



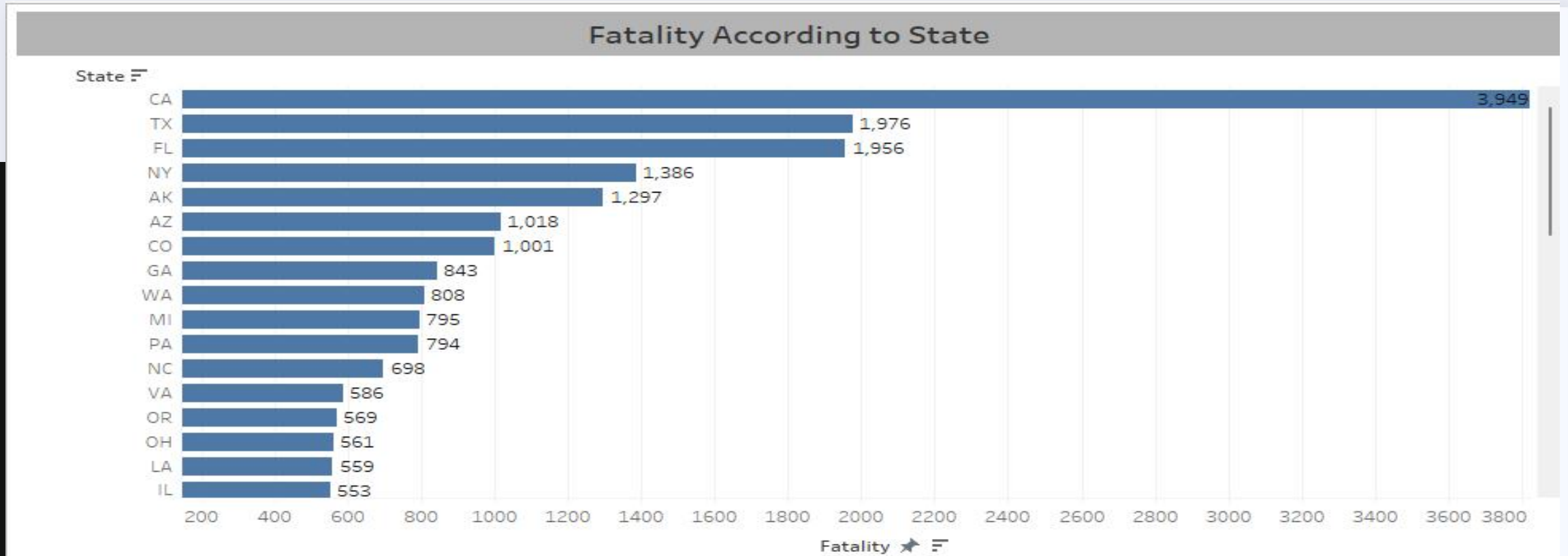
Main Finding:

- Over the past few decades, the number of aviation fatalities has decreased.

Takeaway:

- This is a positive sign, indicating that improvements in technology and safety regulations have had an impact.

Fatalities by Region



Main Finding:

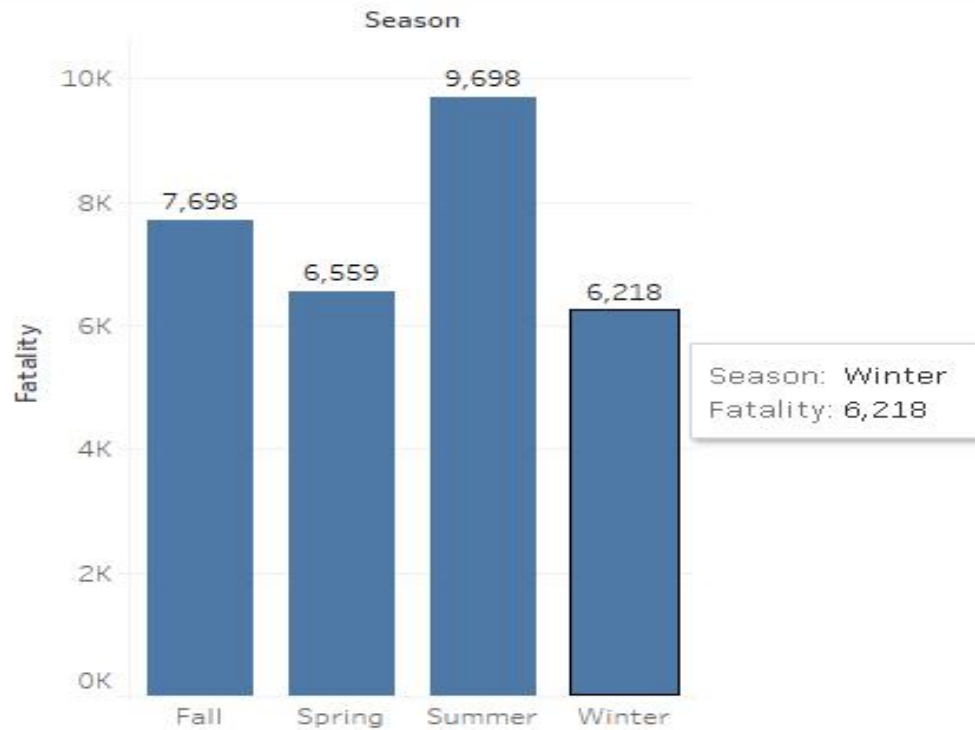
Some regions have a higher number of accidents, particularly in areas with high air traffic and looser safety protocols.

Takeaway:

Regional safety improvements can reduce accidents, especially in higher-risk regions.

Fatalities by Season

Fatality per Season



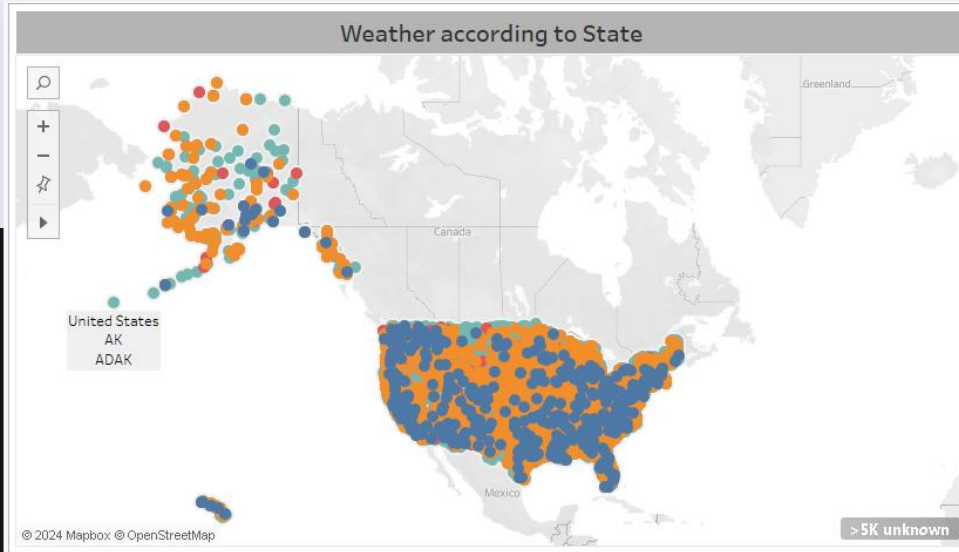
Main Finding:

Summer has the highest fatalities indicating that the model of the airplane may have issues due to hot weather.

Takeaway:

The model components of the planes need to be maintained and invested on to ensure safety of the travellers.

WEATHER, STATE , SEASON AND FATALITIES

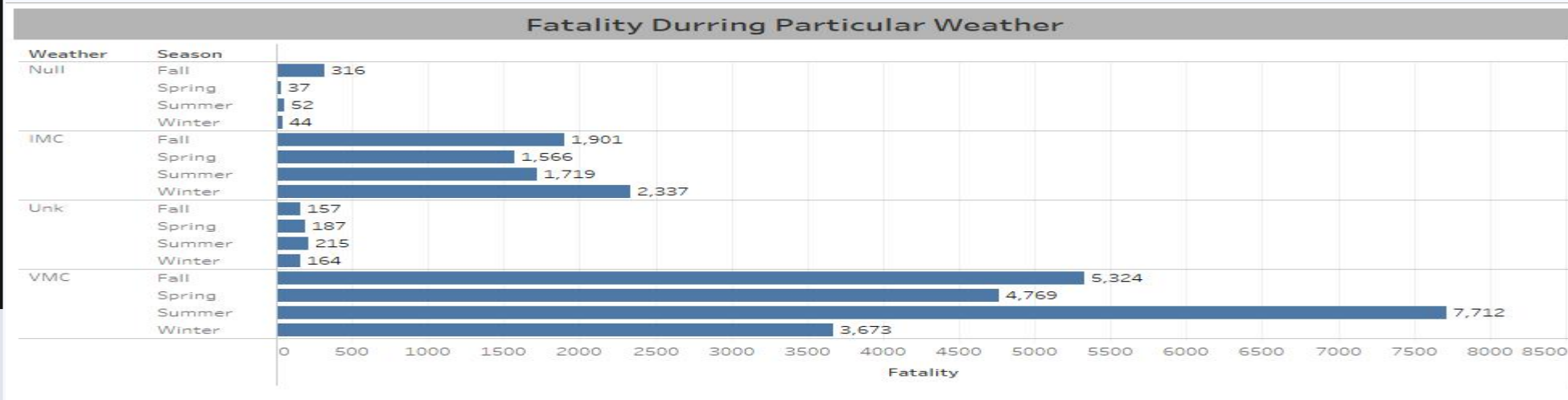


Main Finding:

VMC has the highest number of fatalities indicating that the fault is either with the pilot, personnel to make sure conditions are suitable or aircraft.

Takeaway:

It should be noted about the weather conditions that are safest and most risky to compare findings and come up with solutions on what to do to decrease fatalities.





Actionable Recommendations

Summary of Insights:

- Fatalities are decreasing globally.
- Regions and operators still face risks.
- Critical flight phases like takeoff and landing need extra safety precautions.

Recommendations:

- 1. Regional Safety Audits:** Focus on improving safety in high-risk areas.
- 2. Operator-Specific Strategies:** Develop tailored safety strategies for airlines with higher fatality rates.
- 3. Phase-Specific Safety Measures:** Invest in pilot and crew training, particularly for takeoff and landing.



Conclusion

Key Message:

Aviation safety has improved, but there are still regions, operators, and flight phases that need attention.

Next Steps:

By focusing on high-risk areas and critical phases of flight, we can continue to improve global aviation safety.